What is Amazon S3?

Amazon S3 (Simple Storage Service) is a **cloud-based storage service** provided by AWS (Amazon Web Services).
It allows you to **store and retrieve any amount of data** from anywhere, at any time.

Key points:

- Stores files like images, videos, documents, backups, etc.
- Can be used by individuals, developers, or companies.
- Scales automatically — you don't need to worry about storage limits.
- Provides high durability and availability.

Why use Amazon S3?

Reasons to use S3:

1. **Reliable Storage** – AWS ensures your data is safe and replicated across multiple servers.
2. **Scalable** – You can store from few MBs to petabytes.
3. **Accessible** – Access your files from anywhere via internet.
4. **Secure** – Supports encryption and access control (IAM roles, policies).
5. **Cost-efficient** – Pay only for the storage and services you use.
6. **Integration** – Works with other AWS services like EC2, Lambda, CloudFront.

Terminology to know

- **Bucket:** A container in S3 where all your files (objects) are stored. Think of it like a folder.
- **Object:** A file stored in S3 (like an image, PDF, video).
- **Region:** A geographic area where your bucket is stored. Choose a nearby region for faster access.
- **Key:** The unique name of the object inside a bucket.

How to create a bucket in Amazon S3

Step 1: Login to AWS Management Console

- Go to [AWS Console](#).
- Login using your AWS account credentials.

Step 2: Navigate to S3

- In the search bar, type **S3** and click on it.

Step 3: Click "Create Bucket"

- You will see a button **"Create bucket"**, click it.

Step 4: Configure Bucket

1. **Bucket name:** Give a unique name (e.g., `my-first-bucket-123`).
2. **Region:** Choose a region close to your location.
3. **Settings:**
   - Versioning (optional) – keep multiple versions of files.
   - Public access settings – choose if you want files public or private.
   - Encryption (optional) – enable server-side encryption for security.

Step 5: Review and Create

- Review your settings.
- Click **"Create bucket"**.

Step 6: Upload files

- Open the bucket.
- Click **"Upload"** → select files → click **"Upload"**.

Step 7: Access your files

- Each file will have an **S3 URL** like:
  `https://my-first-bucket-123.s3.amazonaws.com/myfile.jpg`
- Use it to share or access the file.

Public Amazon S3 Vs Private Amazon S3

Step 1: Create a Private S3 Bucket

1. Login to **AWS Management Console** → Go to **S3**.
2. Click **"Create bucket"**.
3. Fill in details:
   - **Bucket name:** `my-private-bucket-123`
   - **Region:** Choose nearest region.
4. **Block all public access:** (keep it private)
5. Click **"Create bucket"**.

Now your bucket is **private by default**, no one can access files yet.

Step 2: Upload an Object to the Bucket

1. Click on your bucket name → **Upload** → **Add files** → Select a file (e.g., `example.txt`).
2. Click **Upload**.

Now your file exists in the **private bucket**.

Step 3: Make the Object Public

1. Click on the object (e.g., `example.txt`) → **Properties** → **Object ownership**.
2. Or, easier: **Select the file** → **Actions** → **Make public**.

Warning: Making it public allows anyone with URL to access the file.

Step 4: Attach a Bucket Policy

To allow access via URL, you can attach a policy. Here's an example policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicReadGetObject",
            "Effect": "Allow",
            "Principal": "*",
            "Action": "s3:GetObject",
            "Resource": "arn:aws:s3:::my-private-bucket-123/*"
        }
    ]
}
```

How to attach policy:

1. Go to your **bucket** → **Permissions** tab → **Bucket Policy** → Paste above JSON.
2. Replace `my-private-bucket-123` with your bucket name.
3. Click **Save**.

This allows **anyone with the file URL** to access objects inside the bucket.

Step 5: Access File via URL

- Object URL format:

```
https://<bucket-name>.s3.<region>.amazonaws.com/<object-name>
```

Example:

```
https://my-private-bucket-123.s3.ap-south-1.amazonaws.com/example.txt
```

- Open this in browser → File should be publicly accessible.

Summary Notes for Students

Step	Action
1	Create private bucket (block public access)
2	Upload object/file to bucket
3	Make object public via properties/actions
4	Attach bucket policy for public `GetObject`
5	Access object using S3 URL

What is Static Website Hosting in S3?

Amazon S3 allows you to **host a static website**, meaning **HTML, CSS, JS, images**, etc., can be served **directly from S3** without a server.

Key points:

- Works for websites that **don't require server-side code** (like PHP, Python backend).
- You can define a **default page** (`index.html`) and an **error page** (`error.html`).
- Files are stored in S3 bucket and accessed via a **public URL**.

Steps to Serve Static Files in S3

Step 1: Create a Bucket

1. Go to **AWS Console → S3 → Create bucket**.
2. Give a **unique bucket name** (e.g., `my-static-website-123`).
3. **Region:** Choose nearest region.
4. **Block all public access:** Uncheck "Block all public access" (needed for public website).
5. Click **Create bucket**.

Important: Bucket name **must match the domain name** if using a custom domain.

Step 2: Upload Static Files

1. Open your bucket → Click **Upload** → Add your files:
    - `index.html` → Home page
    - `style.css` → CSS file
    - `script.js` → JS file
    - Any images
2. Click **Upload**.

Step 3: Enable Static Website Hosting

1. Go to your bucket → **Properties tab → Static website hosting**.
2. Choose **Enable**.
3. Enter:
    - **Index document:** `index.html`
    - **Error document:** `error.html`
4. Save changes.

AWS will give a **Website endpoint URL** like:

```
http://my-static-website-123.s3-website.ap-south-1.amazonaws.com
```

Step 4: Set Bucket Policy to Make Files Public

1. Go to **Permissions → Bucket Policy**.

2. Add the following JSON policy to make all files public:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicReadForWebsite",
            "Effect": "Allow",
            "Principal": "*",
            "Action": "s3:GetObject",
            "Resource": "arn:aws:s3:::my-static-website-123/*"
        }
    ]
}
```

Replace `my-static-website-123` with your bucket name.

3. Click **Save**.

Step 5: Test Your Website

1. Open the **Website endpoint URL** in a browser.
2. You should see your **`index.html` page**.
3. Test a wrong URL like:

```
http://my-static-website-123.s3-website.ap-south-
1.amazonaws.com/wrongpage.html
```

- AWS will automatically show your **`error.html`** page.

6. Optional: Make It HTTPS (Secure)

- By default, S3 static websites use **HTTP**.
- To use **HTTPS**, connect the bucket to **Amazon CloudFront** (CDN) and attach an SSL certificate.